

Paul Pham

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EDUCATION

Washington University in St. Louis – St. Louis, MO

Bachelor of Science in Electrical Engineering

GPA: 3.84

Expected Graduation: May 2029

Minors: Computer Science, Philosophy | Dean's List: Fall 2025, Spring 2026

Relevant Coursework: Introduction to Digital Logic and Computer Design, Differential Equations and Dynamical Systems Modeling, Linear Algebra and Component Analysis, Data Structures and Algorithms

TECHNICAL SKILLS

- **Hardware & Embedded:** KiCad (Schematic Capture & Custom PCB Design), CircuitPython, Verilog, FPGA, Raspberry Pi, ESP32, Soldering & Electrical Instrumentation
- **Robotics & Control:** Inverse Kinematics, PID Control Loops, Computer Vision (OpenCV), ArUco Marker-Based Localization
- **Software & Tools:** MATLAB, Python, Java, Git & GitHub, CAD (Fusion 360), VS Code

PROJECT EXPERIENCE

Autonomous Arm w/ Computer Vision (AACV) | Co-Developer

June 2026 – August 2026

- Co-engineered a 6-DOF robotic arm requiring autonomous object detection with no manual control; built an **OpenCV/ArUco**-based vision pipeline in **Python** that converts camera input into real-world coordinates, calibrated to sub-centimeter accuracy off a single 24.75mm reference marker; managed shared codebase and revisions with a co-developer via **Git/GitHub**.
- Derived and coded an **inverse-kinematics** solver to convert target coordinates into commands for all 6 servo joints in real time, enabling the arm to run 2 fully autonomous modes — rainbow-order stacking and color-to-marker matching — with zero pre-programmed paths.
- Built a two-layer calibration system — per-servo angle mapping plus a 3x3 (9-point) interpolated workspace correction grid — to reduce mechanical and optical placement error across the arm's full range of motion.

Pomodoro Cube | Developer

June 2026 – August 2026

- Designed and hand-soldered a custom **KiCad** PCB for an **ESP32**-based device, CAD-modeling an enclosure and magnetic charging dock to fit every component within the 2.2-inch dimensions of a Rubik's Cube.
- Solved a 4-pin hardware I/O constraint by engineering a resistor-ladder circuit that multiplexes 4 physical buttons onto a single analog input, using distinct voltage thresholds to identify each press.
- Programmed **CircuitPython** firmware as a finite-state machine to run 3 interactive modes — a 15-minute quick timer, a Tamagotchi-style pet, and a productivity tracker — without memory fragmentation on the ESP32's limited RAM.

Autonomous PiCar | Developer

May 2026

- Tuned a **PID** controller in **Python** on a **Raspberry Pi** to stabilize real-time motor speed, then built an **OpenCV** HSV-masking pipeline to drive 3 autonomous behaviors — blue-target seeking, traffic-light response, and hallway cruising — fusing ultrasonic and accelerometer data for collision avoidance.

FPGA-based Simon Says Game | Developer

May 2026

- Designed and synthesized a memory game onto **FPGA** hardware in **SystemVerilog**, implementing a 5-state finite state machine to control gameplay logic and a 4-bit Linear Feedback Shift Register to generate and validate pseudo-random LED sequences.

K-Means Clustering Algorithm | Co-Author

October – December 2025

- Developed a **K-means clustering** algorithm from first principles in **MATLAB** to classify 1,500 handwritten digits, achieving 71.5% accuracy on a held-out set of 200 test digits.

ACTIVITIES & LEADERSHIP

Vietnamese Student Association (VSA) | Outreach Chair

- Serve as Outreach Chair, personally contacting and negotiating with local and global businesses to secure **\$1,000+** in sponsorships for VSA's annual Formal/cultural event, attended by **400+** students.

Lunar New Year Festival Exec Board | Multimedia Chair

- Edited **11** act intros and performances for WashU's annual Lunar New Year Festival show (**650+** attendees), coordinating directly with each act's coordinators to translate their creative vision into final videos.

Additional Involvement: WashU Design, Build, Fly (DBF) – Avionics Team; WashU Badminton Club